

The continuum hypothesis is nonsense

Unless you're doing metaphysics

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PHIL 238W – History of Analytic Philosophy

May 26, 2026

Outline

- 1 Preliminaries: Cardinal arithmetic
- 2 Interlude: fields and polynomials
- 3 ZFC and the continuum hypothesis
- 4 Philosophy

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Cardinality

Definition (*injectivity, surjectivity, bijectivity*)

Let A, B be sets, and let $f : A \rightarrow B$ be a function.

- ① f is *injective* if no two elements of A map to the same element of B .
- ② f is *surjective* if every element of B is mapped to by some element of A .
- ③ f is *bijective* if it is both injective and surjective.

Definition (*cardinal comparisons*)

The *cardinality* of a set A , denoted $|A|$, satisfies the following properties:

- ① If there exists an injective function $f : A \hookrightarrow B$, then $|A| \leq |B|$.
- ② If there exists a surjective function $f : A \twoheadrightarrow B$, then $|A| \geq |B|$.
- ③ If there exists a bijective function $f : A \xrightarrow{\sim} B$, then $|A| = |B|$.

A *cardinal number* is the cardinality of a set.

Power sets and the continuum

Definition (*power set*)

The *power set* of a set A , denoted $\mathcal{P}(A)$, is the set of its subsets:

$$\mathcal{P}(A) = \{S \subseteq A\}.$$

Compare: Wittgenstein's atomic facts vs. states of affairs/propositions.

The power set can be equivalently characterized as the set of $\{0, 1\}$ -valued functions:

$$2^A := \{f: A \rightarrow \{0, 1\}\}.$$

Question

Given a cardinal number $\kappa = |A|$, how can we make it bigger?

Cantor's diagonalization theorem

Cantor's diagonalization theorem

Let A be a set. Then, $|A| < |\mathcal{P}(A)|$.

Proof for A finite. If A is finite, then $|A| = n$ for some finite natural number $n \in \mathbb{N}$. The cardinality of $\mathcal{P}(A)$ is 2^n . Since $n < 2^n$ for all natural numbers, the result follows.

Proof for $A = \mathbb{N}$. Recall that subsets of $\mathbb{N}$ are equivalent to $\{0, 1\}$ -valued sequences. Suppose that there exists a bijection $f: \mathbb{N} \xrightarrow{\sim} \mathcal{P}(\mathbb{N})$.

	0	1	2	3	...
$f(0)$	0	1	0	0	...
$f(1)$	0	0	0	1	...
$f(2)$	1	0	1	0	...
$f(3)$	1	1	0	1	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$
S	1	1	0	0	...

The diagonalization argument produces a new element $S \in \mathcal{P}(\mathbb{N})$ which is not in the image of f : hence f is not a bijection.

New cardinals from old

Question

Given a cardinal number $\kappa = |A|$, how can we make it bigger?

By Cantor's theorem, we can take $2^\kappa := |\mathcal{P}(A)|$, which is guaranteed to be strictly greater than κ . Or:

Definition (*successor cardinal*)

Let $\kappa = |A|$ be a cardinal number. The *successor* of κ , denoted κ^+ , is the smallest cardinal number bigger than κ :

$$\kappa^+ = \min \{ \lambda \text{ cardinal} : \kappa < \lambda \}.$$

For n finite, this is overkill: $n^+ = n + 1 \leq 2^n$.

For κ infinite, addition is insufficient: $\kappa = \kappa + 1$ (Hilbert's hotel).

Hence we have the question: what is the relationship between κ^+ and 2^κ ?

By Cantor's theorem, $\kappa < 2^\kappa \rightarrow \kappa^+ \leq 2^\kappa$. Is this also an equality?

The continuum hypothesis

Proposition

The set of real numbers has the same cardinality as the power set of the natural numbers: $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$.

Therefore, the set theorist's definition of a *real number*: a subset of $\mathbb{N}$.

The cardinality of the natural numbers is denoted $\aleph_0 = |\mathbb{N}|$. Its successor is denoted $\aleph_1 = \aleph_0^+$.

By above, we have that $\aleph_1 = \aleph_0^+ \leq 2^{\aleph_0} = 2^{|\mathbb{N}|} = |\mathcal{P}(\mathbb{N})| = |\mathbb{R}|$.

The continuum hypothesis (CH)

The above inequality is an equality: that is, $\aleph_1 = 2^{\aleph_0} = |\mathbb{R}|$.

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The fundamental theorem of algebra

The fundamental theorem of algebra (FTA)

Every nonconstant polynomial $p(x)$ has at least one root (i.e., solution to the equation $p(x) = 0$).

This is nonsense: we haven't specified what the term *polynomial* means.

Definition (*polynomial*)

A *polynomial with coefficients in a field F* is an expression of the form

$$p(x) = a_0 + a_1x + \cdots + a_nx^n,$$

where $a_0, \dots, a_n \in F$.

A *field* is a set F satisfying the *axioms of field theory*: given two elements in F , we can *add*, *subtract*, *multiply*, and *divide* them (except for division by 0).

Examples: the rationals $\mathbb{Q}$, the reals $\mathbb{R}$, and the complex numbers $\mathbb{C}$.

Nonexamples: the naturals $\mathbb{N} = \{0, 1, 2, \dots\}$; the set of integers $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$.

Independence of FTA from the axioms of field theory

Proposition (*FTA is independent of the axioms of field theory*)

- ① There exists a field such that $\neg$ FTA holds; therefore, FTA is not provable from the axioms of field theory.
- ② There exists a field such that FTA holds; therefore, $\neg$ FTA is not provable from the axioms of field theory.

Proof of (1). FTA is false in $\mathbb{Q}$: the polynomial $x^2 - 2$ has no roots in $\mathbb{Q}$.

Proof of (2). FTA is true in $\mathbb{C}$: this is the (classical) fundamental theorem of algebra.

Motivating Question

Starting with the field $\mathbb{R}$, how could we have come up with the fields $\mathbb{Q}$ and $\mathbb{C}$?

Strategy A: subfield

Let F be a field. A *subfield* $E \subseteq F$ is a subset of F that is also a field.

Example. The rationals can be written as

$$\mathbb{Q} = \{x \in \mathbb{R} : x \text{ is rational}\}.$$

where *rationality* means

$$\text{there exist } p, q \in \mathbb{Z} \text{ s.t. } x = \frac{p}{q}$$

Strategy B: field extensions

Let F be a field. An *extension* is a field K such that $F \subseteq K$.

The most common way to produce field extensions is to (i) adjoin a new element and (ii) force it to obey a certain relation.

Example. The complex numbers $\mathbb{C}$ is an extension of $\mathbb{R}$, given by *adjoining* an element i and *forcing* it to satisfy $i^2 + 1 = 0$.

Adjoining an element i (with no imposed conditions) gives the polynomial ring

$$\mathbb{R}[i] = \{a_0 + a_1i + \cdots + a_ni^n : a_0, \dots, a_n \in \mathbb{R}\}.$$

Forcing $i^2 + 1 = 0$ amounts to taking a *quotient*

$$\mathbb{C} = \mathbb{R}[i]/(i^2 + 1) = \{a + bi : a, b \in \mathbb{R}\}.$$

Technical problems: (1) $\mathbb{R}[i]$ is not a field: it is a more general object called a *ring* (can only add, subtract, and multiply).

(2) The quotient is not always a field: the polynomial $i^2 + 1$ must be *irreducible*.

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CH is independent of the axioms of set theory

A *field* is a set satisfying the *axioms of field theory*.

A *model of set theory* is a class satisfying the *axioms of set theory* (ZFC).

Proposition (*CH is independent of ZFC*)

Assume that ZFC is consistent; i.e., there exists a model of ZFC, which we call the *ground model* V . Then,

- ① (Gödel 1940) There exists a model such that CH holds; therefore, $\neg\text{CH}$ is not provable from ZFC.
- ② (Cohen 1963) There exists a model such that $\neg\text{CH}$ holds; therefore, CH is not provable from ZFC.

Think of V like the field $\mathbb{R}$: we want to start from here to construct (a) a smaller model (like $\mathbb{Q}$) and (b) an extension model (like $\mathbb{C}$).

Strategy A: inner models of ZFC

The idea in (Gödel 1940) is to start with a *ground model* V and construct an *inner model* of *definable sets* called the *constructible hierarchy*:

$$L = \{S \in V : S \text{ is definable}\},$$

where *definability* (roughly) means:

there exists a “not too complicated” formula φ s.t. $S = \{x \in V : \varphi(x)\}$.

We can show that L is a model of set theory, and that in L , $2^{\aleph_0} = \aleph_1$, such that CH holds.

Strategy B: forcing extensions

The idea in (Cohen 1963) is to start with a ground model V and construct an *extension model*: adjoin an object and force it to have certain properties.

Recall: $\neg\text{CH}$ states that $\aleph_1 < 2^{\aleph_0} = |\mathcal{P}(\mathbb{N})|$. Equivalently: $\aleph_2 \leq |\mathcal{P}(\mathbb{N})|$; i.e., there exists an injective function $F: A \hookrightarrow \mathcal{P}(\mathbb{N})$, where $|A| = \aleph_2$.

Sketch of the idea: Adjoin F to our ground model to get an adjoined model $V[F]$. Then we force the condition that F is injective to get a quotient model $V[F]/(F \text{ is injective})$. This model has an injective function $F: A \hookrightarrow \mathcal{P}(\mathbb{N})$, and hence

$$\aleph_2 = |A| \leq |\mathcal{P}(\mathbb{N})| = 2^{\aleph_0}.$$

Therefore $\neg\text{CH}$ holds in $V[F]/(F \text{ is injective})$.

Technical problems: (1) $V[F]$ is not a model of set theory: it has an intuitionistic, not classical, logic (i.e., statements like $x \in A$ do not satisfy the law of the excluded middle).

(2) The quotient of an intuitionistic model is not always a classical model: the condition F is injective must satisfy a *maximality* requirement.

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Platonism and ZFC

*The metamathematical results I have in mind are all centered around...[the incompleteness of mathematics]. This fact is encountered in its simplest form, when the axiomatic method is applied, not to some **hypothetico-deductive system** as geometry (where the mathematician can assert only the conditional truth of the theorems), but **mathematics proper**; i.e., to the body of those mathematical propositions, which hold in an absolute sense without any further hypothesis.*

(Gödel (1951), *Some basic theorems*)

Gödel's position:

- (G1) (Platonism) Mathematical entities exist independently of the human mind, and are perceived (incompletely) by the human mind.
- (G2) The statements of mathematics proper are either true or false (descriptions of the mathematical world).

The independence of CH from ZFC seems to threaten Gödel's platonism. I will argue that there are two distinct ways to retain (G1) and (G2):

- (I) Find new axioms for set theory, in addition to ZFC. (Gödel (1931), p. 476)
- (II) Accept that set theory is not *mathematics proper*, and is instead a *hypothetico-deductive system* like geometry. (Me, (right now))

(I) Gödel's position

Recall: Gödel's *constructible hierarchy* L is a model of set theory where every set is *definable*. Gödel's Axiom A ($V = L$) (see Gödel (1931), *What is Cantor's continuum problem?* p. 478) states that this is the *correct* model of set theory: L is the correct description of the mathematical universe.

Gödel did not like this: $V = L$ implies certain facts in point-set topology that Gödel found unintuitive.

Analogy (Banach–Tarski paradox): Assuming the axiom of choice, one can find a decomposition of the unit ball into 5 pieces, which can be rearranged into two disjoint unit balls. If one finds this to be unintuitive, then one has defeasible reasons to reject the axiom of choice.

The axiom $V = L$ makes the mathematical universe “small”. Gödel wanted axioms which would make the mathematical universe “big”: hence *large cardinal axioms* in modern set theory.

(II) Set theory is a hypothetico-deductive system

The statement $\aleph_1 = 2^{\aleph_0}$ is nonsense. Why? The meanings of these terms are not fixed independently of the model of set theory under discussion.

Recall:

$$\aleph_1 = \aleph_0^+ = \min \{ \lambda \text{ cardinal} : \lambda > \aleph_0 \}.$$

A *cardinal* is the cardinality of a set. Models of set theory tell us what is a set and what is not a set.

Hence: $\aleph_1$ is meaningless on its own; it must receive an interpretation indexed to a model of set theory.

Compare: *polynomial* in FTA is meaningless on its own; it must receive an interpretation indexed to a field which its coefficients are allowed to be.

Given Gödel's platonism (G1) and (G2), this bottoms out to a *multiverse view* of ZFC: there exist many models of ZFC, and we can study them and their properties as we would different models of hypothetico-deductive systems like geometry and algebra.